# TARAN SIRUVURI

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## EDUCATION

**The University of Texas at Austin** **May 2028**

Bachelor of Science, Mechanical Engineering | GPA: 4.00/4.00

Tau Beta Pi Honor Society | Pursuing BDP Certificate in Innovation, Creativity & Entrepreneurship

## ENGINEERING EXPERIENCE

**Throne Science | Research and Development Intern** **Aug 2026 - Present**

Austin, TX

- Redesign robotic arm, extrusion mechanism, and carriage assemblies for a modular sampling robot; revise component layouts and mechanical packaging for space-constrained operation.
- Develop CAD models for sensor mounts, electrical housings, and structural components, addressing mechanical/electrical interfaces, assembly access, and serviceability.

**General Motors | Mechanical Engineering Intern** **Jun 2026 - Aug 2026**

Lansing Delta Township, MI

- Diagnosed automated body-shop equipment failures, including a belt-drive motor issue; cross-checked replacement frame-hoist specifications against OEM drawings to identify part-compatibility risks.
- Isolated robotic alignment errors behind vehicle door-gap deviations; implemented 0.5 mm datum adjustments and a 0.25 mm weldment correction, validating 3 builds and eliminating recurring manual rework.
- Audited 200 motors across 5+ vendors; reconciled installed part numbers, gearbox configurations, equipment locations, and pallet-system associations against maintenance records.
- Standardized motor records and sourcing procedures; reduced projected diagnostic and replacement-planning time from 6-8 hours to under 3 hours, an expected reduction exceeding 50%.

**Austin Reliability Labs | Reliability Engineering Intern** **Jun 2025 - Aug 2025**

Austin, TX

- Designed and fabricated grounding, mounting, and alignment fixtures for 15 Dell motherboards, establishing repeatable hardware setups for HALT and environmental testing.
- Improved test-setup repeatability by 30% through fixture and procedure changes; developed a fiber-optic test fixture recognized with an Innovator of the Month award.
- Executed vibration, shock, thermal cycling, humidity, and power-cycling tests; tracked hardware failures and supported root-cause investigations under accelerated stress conditions.

## ENGINEERING PROJECTS

**Longhorn Racing Electric | Chassis Torsional Rigidity** **Sep 2025 - Mar 2026**

- Built ANSYS chassis torsion models and compared predicted deflection with physical-test results to investigate simulation-to-test discrepancies.
- Evaluated critical load paths, tube placement, and battery packaging to guide chassis geometry revisions and stiffness/weight tradeoffs.

**Longhorn Racing Electric | Chassis Hardpoint Fixture** **Sep 2024 - Mar 2025**

- Designed SolidWorks hardpoint fixtures using GD&T and tolerance-stack analysis to locate critical chassis nodes within 0.005 inch.
- Machined and TIG tack-welded fixture subassemblies; established repeatable assembly and welding procedures for vehicle construction.

**3D Printed Ventilation Grille | Electronics Cooling**

- Designed a slotted ventilation grille with corner standoffs for electronics cooling and 3D printed it in BASF Forward AM Ultrafuse PC/ABS FR, a flame-retardant filament suited to high temperatures.

## TECHNICAL SKILLS

**Test & Validation:** HALT/HASS, vibration, shock, thermal cycling, humidity, power cycling, failure tracking, root-cause analysis, test procedures

**Mechanical & Robotics:** SolidWorks, Inventor, ANSYS finite element analysis (FEA), GD&T, tolerance stack-up, fixture/tooling design, electromechanical packaging, mechanical assembly

**Data & Fabrication:** Python, MATLAB, Excel, test documentation, CNC machining, mill, lathe, 3D printing, TIG welding, design for manufacturability (DFM)